Enhanced TCP WestwoodNew with Proportional Congestion Decay and Adaptive RTO Estimation

Md. Afham Adian

Student ID: 2105119

Base Paper:

Enhanced TCP Westwood Congestion Avoidance Mechanism (TCP WestwoodNew)

Background and Motivation

TCP was originally designed for wired networks, where packet loss is a reliable indicator of congestion. On wireless networks, however, packets drop for other reasons (signal interference, handoffs, transient channel degradation), making the standard TCP assumption fundamentally flawed.

TCP Westwood introduced Bandwidth Estimation (BWE): it monitors the rate at which acknowledgments return and uses that to estimate available bandwidth, producing a more accurate reaction to loss events on wireless links.

TCP WestwoodNew extends this with two targeted improvements:

1.1 Smarter Congestion Avoidance

Original Westwood increments `cwnd` (congestion window) by $\frac{1}{\texttt{cwnd}}$ on every ACK regardless of what the network is actually doing. WestwoodNew introduces a bandwidth ratio check before deciding how aggressively to grow:

$$\text{BWratio} = \frac{BW_{\text{new}}}{BW_{\text{prev}}}$$

- $\text{BWratio} \geq 1.5 \quad \Rightarrow \quad$ network is improving $\Rightarrow$ grow faster: $+\frac{2}{\texttt{cwnd}}$
- $1 \leq \text{BWratio} < 1.5 \quad \Rightarrow \quad$ network is stable $\Rightarrow$ grow normally: $+\frac{1}{\texttt{cwnd}}$
- $\text{BWratio} < 1 \quad \Rightarrow \quad$ network load increasing $\Rightarrow$ hold: $+0$

The algorithm reacts to real-time conditions rather than blindly following a fixed growth schedule.

1.2 Better RTO Estimation

The standard RTO miscalibration problem cuts both ways: waiting too long wastes throughput, retransmitting too early (spurious RTO) floods the network and triggers unnecessary slow starts. WestwoodNew computes a route-aware RTO:

$$\mathrm{RTO}_{\mathrm{new}} = \frac{\mathrm{RTT}_{\mathrm{new}}}{\mathrm{RTT}_{\mathrm{old}}} \times \mathrm{RTO}_{\mathrm{old}}$$

If the route changes and RTT jumps, RTO scales proportionally. This is especially relevant on wireless links where route changes are common.

Project Objective

The objective of this project is to design, implement, and evaluate an enhanced version of TCP WestwoodNew that addresses two specific limitations of the original algorithm. The enhanced protocol is intended to deliver more stable and responsive congestion control, particularly in wireless and variable-delay network environments.

Proposed Modifications

3.1 Modification 1: Proportional Congestion Decay

Problem. When BWratio < 1, WestwoodNew freezes `cwnd`. This is too passive. The sender keeps injecting the same volume of traffic while conditions are actively worsening, which allows queues to build and exacerbates delay and eventual loss.

Proposed fix. Replace the freeze with a proportional decay:

$$\texttt{cwnd} = \max\Big(\texttt{cwnd} \times \big(1 - \gamma(1 - \mathrm{BWratio})\big),\ 1\Big)$$

where $\gamma \in (0, 1]$ is a tunable sensitivity parameter. The reduction in `cwnd` is directly proportional to how far the bandwidth ratio has dropped below 1. This produces a smooth, continuous response rather than a binary freeze, allowing the sender to back off progressively as conditions deteriorate.

3.2 Modification 2: Smoothed Adaptive RTO Estimation

Problem. The original RTO update $\mathrm{RTO}_{\mathrm{new}} = \frac{\mathrm{RTT}_{\mathrm{new}}}{\mathrm{RTT}_{\mathrm{old}}} \times \mathrm{RTO}_{\mathrm{old}}$ reacts directly to each raw RTT sample with no smoothing. A single transient delay spike (common on wireless links) produces an inflated RTO and may trigger a false slow start, degrading throughput unnecessarily.

Proposed fix. Apply a weighted moving average before feeding RTT into the RTO update:

$$\mathrm{RTT}_{\text{smooth}} = \alpha \cdot \mathrm{RTT}_{\text{new}} + (1-\alpha) \cdot \mathrm{RTT}_{\text{smooth, prev}}$$

$$\mathrm{RTO}_{\text{new}} = \frac{\mathrm{RTT}_{\text{smooth}}}{\mathrm{RTT}_{\text{old}}} \times \mathrm{RTO}_{\text{old}}$$

where $\alpha \in (0,1)$ controls the responsiveness-stability tradeoff. Using $\mathrm{RTT}_{\text{smooth}}$ in place of the raw $\mathrm{RTT}_{\text{new}}$ filters out isolated noise while still tracking genuine path changes. The result is a more stable RTO on links with bursty or variable delay, without sacrificing meaningful responsiveness.

Expected Outcomes

4.1 Proportional Decay

By reducing `cwnd` in proportion to the bandwidth drop rather than holding it constant, the sender reacts earlier and more smoothly to deteriorating conditions. This should translate directly to reduced queue buildup, lower end-to-end delay, and fewer packet drops in scenarios where network conditions degrade gradually. The binary freeze in WestwoodNew provides no congestion relief during the worsening window; the proportional decay closes that gap.

4.2 Smoothed RTO

Filtering transient RTT spikes before they influence RTO prevents false slow starts triggered by isolated noisy samples. On links with variable or bursty delay (typical of wireless environments) this should produce more stable throughput and fewer spurious retransmissions. The core sensitivity to genuine route changes is preserved; only isolated transients are attenuated.

4.3 Combined Effect

Together, the two modifications are expected to produce a measurably more robust and responsive protocol than the original WestwoodNew, with clear improvements in throughput, end-to-end delay, and packet loss rate across variable and wireless network conditions.